

GED Science Practice Test 10

PART A — QUESTIONS

GED-SCI10-001

Section: Science

Type: mcq

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student investigates how different light schedules affect the flowering of a short-day plant. Four groups of identical plants are kept under the same temperature, water, and soil conditions. Only the number of hours of darkness changes each day. After three weeks, the student counts the plants that flower.

Hours of darkness per day | Plants that flower out of 20

6		1
8		3
10		9
12		17

Which conclusion is best supported by the data?

- A) More hours of darkness were associated with more flowering.
- B) Plants flowered most with 6 hours of darkness.
- C) Darkness had no relationship to flowering.
- D) All plants flowered under every light schedule.

GED-SCI10-002

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

Kinetic energy can be calculated as $\frac{1}{2} \times \text{mass} \times \text{speed}^2$.

Object | Mass (kg) | Speed (m/s)

A		2		4
B		4		3
C		6		2
D		8		1

Which object has the greatest kinetic energy?

- A) Object A
- B) Object B
- C) Object C
- D) Object D

GED-SCI10-003

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

A coastal community uses seawalls, sand dunes, and restored marshes to reduce damage from storm surge. Marshes can slow moving water, trap sediment, and provide habitat. Dunes can absorb wave energy before water reaches buildings.

Which statement best describes how these features help the community?

- A) They can reduce storm-surge impact by slowing or absorbing moving water.
- B) They stop the Moon from affecting tides.
- C) They prevent all future storms from forming.
- D) They remove all salt from ocean water.

GED-SCI10-004

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

Bone tissue is living tissue. Bone cells help repair small cracks, and minerals stored in bone help make the skeleton strong. Bone marrow inside some bones produces blood cells.

Which statement best describes one function of the skeletal system?

- A) It supports the body and helps produce blood cells.
- B) It pumps air into the lungs.
- C) It filters urine from the blood.
- D) It changes sunlight into sugar.

GED-SCI10-005

Section: Science

Type: numeric_entry

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A force of 18 N acts on an object and causes an acceleration of 3 m/s².

Force = mass × acceleration

What is the mass of the object in kilograms?

Type your answer as a whole number.

GED-SCI10-006

Section: Science

Type: multi_select

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Select TWO answers.

A student tests how the thickness of a rubber band affects the distance a paper airplane launcher sends a paper airplane. The same airplane is used in every trial, and the rubber band is pulled back the same distance each time.

Which TWO factors should be kept constant?

- A) Same paper airplane
- B) Pullback distance
- C) Rubber band thickness
- D) Distance traveled
- E) Final measured result

GED-SCI10-007

Section: Science

Type: dropdown

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

During protein synthesis, DNA instructions are copied into messenger RNA. The messenger RNA then carries the instructions to ribosomes, where amino acids are joined to form [D1]. This process helps explain how genes can affect [D2].

Dropdowns:

D1) proteins | rocks | water vapor

D2) traits | ocean tides | mineral hardness

GED-SCI10-008

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A student inflates a bicycle tire. The air particles inside the tire collide with the tire walls. As more air is pumped into the same tire volume, the number of particles inside increases.

Which statement best explains why tire pressure increases?

- A) More gas particles collide with the tire walls.
- B) Gas particles stop moving inside the tire.
- C) Air changes into a solid metal.
- D) The tire produces gravity.

GED-SCI10-009

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

A weather station records relative humidity and afternoon thunderstorm reports.

Day | Relative humidity (%) | Thunderstorm reported?

1 | 42 | No

2 | 55 | No

3 | 71 | Yes

4 | 84 | Yes

Which conclusion is best supported by the data?

- A) Thunderstorms were reported on the higher-humidity days.

- B) Thunderstorms were reported only on the lowest-humidity days.
- C) Relative humidity stayed the same every day.
- D) Humidity prevented clouds from forming.

GED-SCI10-010

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A researcher records the amount of glycogen stored in muscle cells before and after a long-distance run.

Condition | Muscle glycogen (g)

Before run | 410

After run | 170

Which statement is best supported by the data?

- A) Muscle glycogen decreased after the long-distance run.
- B) Muscle glycogen increased after the long-distance run.
- C) Muscle glycogen stayed exactly the same after the run.
- D) Muscle cells contained no glycogen before the run.

GED-SCI10-011

Section: Science

Type: drag_drop

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Match each target with the correct tile.

Tiles:

- A) Mass attached to the spring scale
- B) Amount the spring stretches
- C) Same spring scale used in every trial
- D) Color of the lab table

Targets:

- T1) Independent variable
- T2) Dependent variable
- T3) Controlled variable

GED-SCI10-012

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A synapse is a small gap between neurons. When an electrical impulse reaches the end of one neuron, chemicals called neurotransmitters are released. These chemicals cross the synapse and can trigger a response in the next neuron.

Which statement best describes the role of neurotransmitters?

- A) They carry signals across synapses between neurons.
- B) They form sedimentary rocks.
- C) They store water in plant vacuoles.
- D) They reflect light from a mirror.

GED-SCI10-013

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A student charges a plastic comb by rubbing it with a cloth. The comb then attracts small neutral paper pieces. The charged comb causes charges inside the paper pieces to shift slightly, making the side near the comb oppositely charged.

Which idea best explains the attraction?

- A) Electric charges can attract after charge separation in the paper.
- B) The paper pieces become tiny planets.
- C) The comb removes all mass from the paper.
- D) Electric forces act only in water.

GED-SCI10-014

Section: Science

Type: numeric_entry

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

A weather balloon rises from 500 meters to 2,900 meters in 40 minutes.

What is the average rise rate in meters per minute?

Type your answer as a whole number.

GED-SCI10-015

Section: Science

Type: mcq

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A researcher studies whether a new physical therapy routine improves knee flexibility after injury. One group follows the new routine, and a similar group follows the standard routine. Both groups exercise for 20 minutes per day. Knee flexibility is measured after six weeks.

Why is the standard-routine group included?

- A) It provides a comparison for judging the new routine.
- B) It guarantees that no one improves.
- C) It changes knee tissue into bone.
- D) It removes the need to measure flexibility.

GED-SCI10-016

Section: Science

Type: mcq

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student compares how different string tensions affect the pitch of a simple string instrument. The student changes the tension but keeps the string length and string material the same. The frequency of the sound is measured.

What is the dependent variable?

A) Frequency of the sound

B) String tension

C) String length

D) String material

GED-SCI10-017

Section: Science

Type: dropdown

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

An object at rest tends to stay at rest unless acted on by a net force. An object in motion tends to continue moving at the same velocity unless acted on by a net force. This tendency is called [D1], and it is related to an object's [D2].

Dropdowns:

D1) inertia | evaporation | digestion

D2) mass | color | temperature only

GED-SCI10-018

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A clinic records average breathing rate before and after a mild asthma treatment.

Condition | Average breathing rate (breaths/min)

Before treatment | 26

After treatment | 18

Which statement is best supported by the data?

A) Average breathing rate decreased after treatment.

B) Average breathing rate increased after treatment.

C) Breathing rate stayed exactly the same after treatment.

D) No patients were breathing before treatment.

GED-SCI10-019

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student records how far a block slides after being pushed across the same surface with different starting speeds.

Starting speed (m/s): 1, 2, 3, 4

Sliding distance (m): 0.4, 1.6, 3.6, 6.4

Which statement is best supported by the data?

- A) Higher starting speed is associated with greater sliding distance.
- B) Higher starting speed is associated with shorter sliding distance.
- C) Sliding distance stays the same for all starting speeds.
- D) The block never moves when pushed.

GED-SCI10-020

Section: Science

Type: multi_select

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

Select TWO answers.

A town compares two land areas after a heat wave.

Land area | Tree cover (%) | Average afternoon air temperature (°C)

Area A | 12 | 38

Area B | 48 | 32

Which TWO statements are best supported by the data?

- A) Area B had more tree cover than Area A.
- B) Area B had a lower average afternoon air temperature than Area A.
- C) Area A had more tree cover than Area B.
- D) Area A had the lower average afternoon air temperature.
- E) Both areas had exactly the same temperature.

GED-SCI10-021

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

The lymphatic system includes lymph vessels, lymph nodes, and white blood cells. Lymph nodes can trap pathogens and help activate immune cells. The lymphatic system also returns excess fluid from tissues back to the bloodstream.

Which function is described?

- A) Immune defense and fluid return
- B) Production of sunlight by plants
- C) Movement of tectonic plates
- D) Reflection of sound waves only

GED-SCI10-022

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A student compares two forces acting on a book on a table. Gravity pulls the book downward. The table pushes upward with an equal force, so the book does not accelerate.

Which statement best describes the forces on the book?

- A) The forces are balanced, so the book remains at rest.
- B) The forces are unbalanced, so the book must speed up.
- C) Gravity stops acting on the book.
- D) The table force turns into light energy.

GED-SCI10-023

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

A model describes how a solar eclipse occurs. The Moon moves between Earth and the Sun. The Moon blocks sunlight from reaching a small part of Earth's surface.

Which arrangement causes a solar eclipse?

- A) Sun, Moon, Earth
- B) Moon, Earth, Sun
- C) Earth, Sun, Moon
- D) Earth, Mars, Moon

GED-SCI10-024

Section: Science

Type: numeric_entry

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A group of 150 people is screened for high blood pressure. Of these people, 39 have high blood pressure.

What percent of the people have high blood pressure?

Type your answer as a percent rounded to the nearest whole number.

GED-SCI10-025

Section: Science

Type: mcq

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student wants to test how wheel size affects the distance a model vehicle travels after rolling

down a ramp. The student changes wheel size but keeps the ramp height, vehicle mass, and starting position the same.

Which factor is the independent variable?

- A) Wheel size
- B) Distance traveled
- C) Ramp height
- D) Vehicle mass

GED-SCI10-026

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A mutation changes one base in a gene, causing one amino acid in a protein to be different. The changed protein folds incorrectly and cannot perform its normal job in the cell.

Which statement best explains the effect of the mutation?

- A) A DNA change can alter protein structure and function.
- B) A DNA change always improves protein function.
- C) Proteins do not depend on genetic instructions.
- D) Amino acids control the orbit of the Moon.

GED-SCI10-027

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student measures the output of a small solar cell at different light intensities.

Light intensity (units): 100, 200, 300, 400

Current produced (mA): 12, 24, 36, 48

Which statement best describes the relationship?

- A) Current increased in direct proportion to light intensity.
- B) Current decreased as light intensity increased.
- C) Current stayed the same at every light intensity.
- D) Light intensity and current had an opposite relationship.

GED-SCI10-028

Section: Science

Type: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student compares the average distance of four objects from Earth.

Object | Average distance from Earth

Low-orbit satellite | 400 km

Moon | 384,000 km

Mars at close approach | 56,000,000 km
Sun | 150,000,000 km

Which object listed is closest to Earth?

- A) Low-orbit satellite
- B) Moon
- C) Mars at close approach
- D) Sun

GED-SCI10-029

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A researcher records the average reaction time for four age groups during a simple computer test.

Age group | Average reaction time (milliseconds)

Teens | 240

Young adults | 215

Middle-aged adults | 260

Older adults | 330

Which age group had the fastest average reaction time?

- A) Teens
- B) Young adults
- C) Middle-aged adults
- D) Older adults

GED-SCI10-030

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A firefighter uses a foam extinguisher on a small fuel fire. The foam covers the fuel surface and reduces contact between the fuel and oxygen in the air.

Which part of the fire triangle is mainly reduced?

- A) Oxygen
- B) Gravity
- C) Sound
- D) Magnetism

GED-SCI10-031

Section: Science

Type: multi_select

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Select TWO answers.

A student studies how different saltwater concentrations affect the hatching rate of brine shrimp eggs. The student places equal numbers of eggs into solutions with different salt concentrations and counts hatched shrimp after 48 hours.

Which TWO steps would improve the investigation?

- A) Use the same number of eggs in each solution.
- B) Keep the hatching time the same for all groups.
- C) Change salt concentration and temperature together.
- D) Count only the container with the most hatched shrimp.
- E) Use a different egg type in each solution.

GED-SCI10-032

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student stretches a spring with different forces.

Force (N): 3, 6, 9, 12

Stretch (cm): 2, 4, 6, 8

How much does the spring stretch when the force is 9 N?

- A) 2 cm
- B) 4 cm
- C) 6 cm
- D) 8 cm

GED-SCI10-033

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

Barnacles often attach to whales. The barnacles gain access to moving water and food particles. The whale is usually not helped or harmed significantly by the barnacles.

Which relationship is described?

- A) Commensalism
- B) Predation
- C) Parasitism
- D) Competition

GED-SCI10-034

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

The pancreas releases digestive enzymes into the small intestine. These enzymes help break down carbohydrates, proteins, and fats into smaller molecules that can be absorbed.

Which statement best describes this function of the pancreas?

- A) It produces enzymes that help digest food molecules.
- B) It exchanges gases between air and blood.
- C) It forms the hard outer layer of teeth.
- D) It stores most genetic information in neurons.

PART B — ANSWER KEY + EXPLANATIONS

GED-SCI10-001 — Correct: A — Correct Answer: More hours of darkness were associated with more flowering. — Explanation: The number of flowering plants increased from 1 to 17 as hours of darkness increased.

GED-SCI10-002 — Correct: B — Correct Answer: Object B — Explanation: Kinetic energy = $\frac{1}{2} \times \text{mass} \times \text{speed}^2$. Object B has $\frac{1}{2} \times 4 \times 3^2 = 18$ J, the greatest value listed.

GED-SCI10-003 — Correct: A — Correct Answer: They can reduce storm-surge impact by slowing or absorbing moving water. — Explanation: Marshes slow water and dunes absorb wave energy, reducing storm-surge effects.

GED-SCI10-004 — Correct: A — Correct Answer: It supports the body and helps produce blood cells. — Explanation: Bones support the body, and marrow inside some bones produces blood cells.

GED-SCI10-005 — Correct: 6 — Tolerance: exact — Correct Answer: 6 — Explanation: Mass = force $\div$ acceleration = $18 \text{ N} \div 3 \text{ m/s}^2 = 6 \text{ kg}$.

GED-SCI10-006 — Correct: A,B — Correct Answer: Same paper airplane | Pullback distance — Explanation: These factors should stay constant so rubber band thickness is the main variable being tested.

GED-SCI10-007 — Correct: D1=proteins; D2=traits — Correct Answer: D1=proteins; D2=traits — Explanation: Genes provide instructions for making proteins, and proteins influence many traits.

GED-SCI10-008 — Correct: A — Correct Answer: More gas particles collide with the tire walls. — Explanation: Adding more air increases the number of particles colliding with the tire walls, raising pressure.

GED-SCI10-009 — Correct: A — Correct Answer: Thunderstorms were reported on the higher-humidity days. — Explanation: Thunderstorms were reported on Days 3 and 4, when humidity was 71% and 84%.

GED-SCI10-010 — Correct: A — Correct Answer: Muscle glycogen decreased after the long-distance run. — Explanation: Glycogen decreased from 410 g to 170 g after the run.

GED-SCI10-011 — Correct: T1=A; T2=B; T3=C — Correct Answer: T1=Mass attached to the spring scale; T2=Amount the spring stretches; T3=Same spring scale used in every trial — Explanation: The mass is changed, the stretch is measured, and the spring scale is controlled.

GED-SCI10-012 — Correct: A — Correct Answer: They carry signals across synapses between neurons. — Explanation: Neurotransmitters cross synapses and pass signals from one neuron to another.

GED-SCI10-013 — Correct: A — Correct Answer: Electric charges can attract after charge separation in the paper. — Explanation: A charged comb can shift charges in neutral paper, causing attraction.

GED-SCI10-014 — Correct: 60 — Tolerance: exact — Correct Answer: 60 — Explanation: The balloon rises 2,400 meters in 40 minutes. $2,400 \div 40 = 60$ m/min.

GED-SCI10-015 — Correct: A — Correct Answer: It provides a comparison for judging the new routine. — Explanation: The standard-routine group allows researchers to compare results with the new-routine group.

GED-SCI10-016 — Correct: A — Correct Answer: Frequency of the sound — Explanation: The dependent variable is the measured frequency after changing string tension.

GED-SCI10-017 — Correct: D1=inertia; D2=mass — Correct Answer: D1=inertia; D2=mass — Explanation: Inertia is the tendency to resist changes in motion, and more mass means more inertia.

GED-SCI10-018 — Correct: A — Correct Answer: Average breathing rate decreased after treatment. — Explanation: The average breathing rate decreased from 26 to 18 breaths/min.

GED-SCI10-019 — Correct: A — Correct Answer: Higher starting speed is associated with greater sliding distance. — Explanation: Sliding distance increased from 0.4 m to 6.4 m as starting speed increased.

GED-SCI10-020 — Correct: A,B — Correct Answer: Area B had more tree cover than Area A. | Area B had a lower average afternoon air temperature than Area A. — Explanation: Area B had 48% tree cover and 32°C, while Area A had 12% tree cover and 38°C.

GED-SCI10-021 — Correct: A — Correct Answer: Immune defense and fluid return — Explanation: Lymph nodes support immune defense, and lymph vessels return excess tissue fluid to the bloodstream.

GED-SCI10-022 — Correct: A — Correct Answer: The forces are balanced, so the book remains at rest. — Explanation: Equal upward and downward forces result in no acceleration.

GED-SCI10-023 — Correct: A — Correct Answer: Sun, Moon, Earth — Explanation: A solar eclipse occurs when the Moon is between the Sun and Earth.

GED-SCI10-024 — Correct: 26 — Tolerance: exact — Correct Answer: 26 — Explanation: $39 \div 150 \times 100 = 26\%$.

GED-SCI10-025 — Correct: A — Correct Answer: Wheel size — Explanation: The independent variable is the factor intentionally changed by the student.

GED-SCI10-026 — Correct: A — Correct Answer: A DNA change can alter protein structure and function. — Explanation: A change in a gene can change an amino acid sequence, affecting protein folding and function.

GED-SCI10-027 — Correct: A — Correct Answer: Current increased in direct proportion to light intensity. — Explanation: Current doubled when light intensity doubled, showing a direct proportional relationship.

GED-SCI10-028 — Correct: A — Correct Answer: Low-orbit satellite — Explanation: The low-orbit satellite is 400 km from Earth, less than the distances listed for the Moon, Mars, and the Sun.

GED-SCI10-029 — Correct: B — Correct Answer: Young adults — Explanation: The fastest reaction time is the lowest time, 215 milliseconds.

GED-SCI10-030 — Correct: A — Correct Answer: Oxygen — Explanation: Foam limits contact between fuel and oxygen, reducing one part of the fire triangle.

GED-SCI10-031 — Correct: A,B — Correct Answer: Use the same number of eggs in each solution. | Keep the hatching time the same for all groups. — Explanation: Keeping egg number and hatching time constant improves the investigation.

GED-SCI10-032 — Correct: C — Correct Answer: 6 cm — Explanation: The table shows that a 9 N force stretches the spring 6 cm.

GED-SCI10-033 — Correct: A — Correct Answer: Commensalism — Explanation: Commensalism occurs when one organism benefits and the other is not significantly helped or harmed.

GED-SCI10-034 — Correct: A — Correct Answer: It produces enzymes that help digest food molecules. — Explanation: The pancreas releases enzymes into the small intestine to help digest carbohydrates, proteins, and fats.